Statistical Inference and Modeling

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Lectures

- Lecture 1: Introduction
- Lecture 2: Model selection
- Lecture 3: Resampling
- Lecture 4: Linear regression
- Lecture 5: Robust regression
- Lecture 6: Classification
- Lecture 7: Logistic regression
- Lecture 8: Neural networks
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- Lecture 10: Bayesian decision, LDA, QDA, and naive Bayes
- Lecture 11: Support vector machine
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- Lecture 13: Bagging, random forecast, boosting
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- Lecture 16: Gaussian mixture model clustering
- Lecture 17: DBSCAN clustering

10. K-nearest neighbor

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K-nearest neighbor —the algorithm

* No model parameter

* 1 Hyper parameter.

radius (กำหนดขอบเขตที่หา) → จะเจอบ่อยใน Paper.

max. neighbors (อิงจาก K ตัวที่ใกล้สุด)

sklearn ใช้มัน.

Setting: Let $\{\boldsymbol{x}_j, y_j\}_{j=1}^{m}$ be the set of training input features and labels.

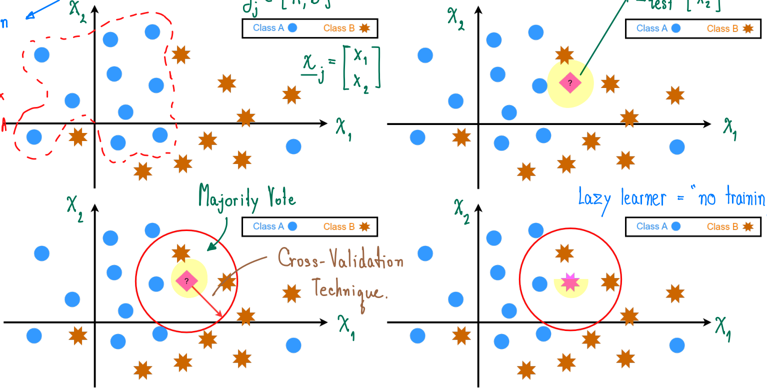
$x_j \in \mathbb{R}^n$

$y_j \in \{A, B\}$

$\underline{x}_j = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$\underline{x}_{test} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$\mu^x_{y=A}, \sigma^{2,x}_{y=A}$



- Find the distance between the input feature and the training features.
- Identify K nearest training features Let Neigh$\{\boldsymbol{x}_{\text{Test}}\}$ be an index set of the neighborhoods of $\boldsymbol{x}_{\text{Test}}$. The prediction is the average of all training labels in the neighborhood set of $\boldsymbol{x}_{\text{Test}}$:

Regression Tasks

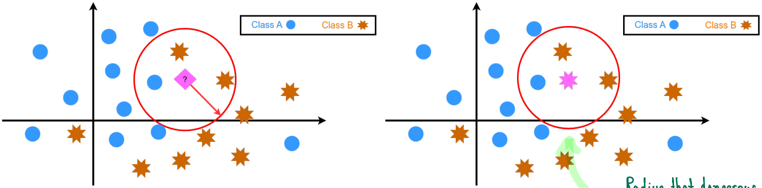
$$\hat{y} = f(\boldsymbol{x}_{\text{Test}}; K) = \frac{1}{K} \sum_{j \in \text{Neigh}\{\boldsymbol{x}_{\text{Test}}\}} y_j$$

Classification

$\hat{y} = \text{Mode}\{y_j \text{ for } j \in \text{Nei}\{x_{test}\}\}$

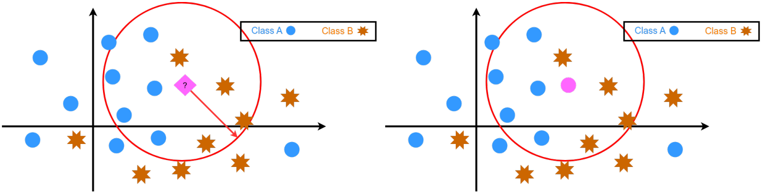
Setting: Let $\{\boldsymbol{x}_j, y_j\}_{j=1}^{m}$ be the set of training input features and labels.

Small radius



Radius that dangerous
→ different result.

Large radius



K-nearest neighbor — probabilistic viewpoint

The last step of finding the class associated with the majority of the K nearest training features (neighbors) could be done in the following steps:

1. count the number of neighbors belonging to each class.
2. compute the ratio between the counted number and K

Remind : Scale data ให้ Range ใกล้เคียงกันด้วย เพราะ ไม่งั้น distance จะมีปัญหา

This is equivalent to estimating the conditional probability:

X_{test}

discriminative tasks

$$p(Y = c|\boldsymbol{x}) = \frac{\#\text{points from } c\text{th class}}{K}$$

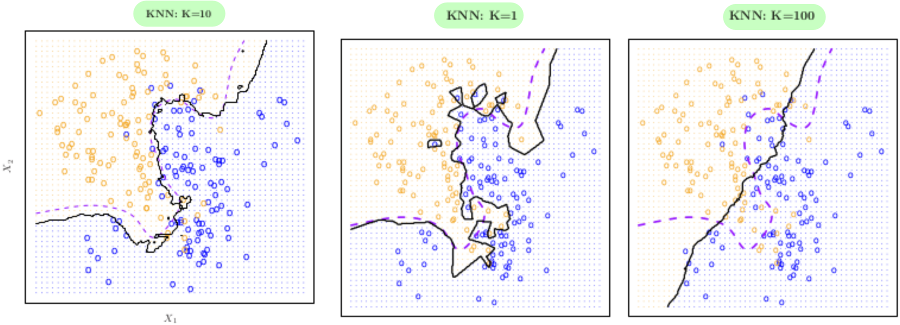
ตอนนี้เราใช้ K เป็น Hyperparameter.

The above Bayes rule is used in KNN to identify the class.

- In KNN, the class with the highest conditional probability will be identified as the class that the testing point $\boldsymbol{x}$ belongs to...
- The optimal value of K depends on **the bias-variance trade-off**. Small K provides the most flexible fit which has low bias but high variance.
- We can choose K from a plot of cross-validated MSE versus $\frac{1}{K}$.
- Note that the scale of variable matters because kNN detects the distance between observations (the variable with a larger scale affects more).
- It is advised to standardize data to have zero mean and unit variance.

Effect of K in K-nearest neighbor

ใช้ Validation Set



— *straight line is KNN, - - - dashed line is Bayes decision.*

- $K = 10$: kNN and Bayes decision boundaries are similar.
- $K = 1$: kNN boundary finds pattern in the data, and is **very flexible** with low bias — it is quite different from Bayes boundary
- $K = 100$: kNN is **less flexible** and is close to linear.
- K is generally taken to be an odd number to minimize ties.

References I

[1] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning*. Springer Series in Statistics. New York, NY, USA: Springer New York Inc., 2001.

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